ChE 377: Momentum Transfer
***Tentative* Course Schedule and Reading Assignments – Spring 2026**

Purdue University ChE 37700: Momentum Transfer, Spring 2026

Course Information

- **ChE 37700 – Momentum Transfer**
- **Instructional Modality** – Face-to-Face
- **Course credit hours –** 4.0
- **Prerequisites:** Undergraduate level CHE 21100 Minimum Grade of C- and (Undergraduate level MA 26600 Minimum Grade of C- [may be taken concurrently] or Undergraduate level MA 30300 Minimum Grade of C- [may be taken concurrently] or Undergraduate level MA 36600 Minimum Grade of C- [may be taken concurrently] or Undergraduate level MA 51400 Minimum Grade of C- [may be taken concurrently] or Undergraduate level ME 58100 Minimum Grade of D- [may be taken concurrently])

Course Description

Catalog description: Differential (microscopic) and integral (macroscopic) mass, momentum, and energy balances. Newtonian and non-Newtonian fluids. Fluid statics. One-dimensional steady and transient laminar flows. Turbulence. Dimensional analysis and similarity. Friction factors and drag coefficients. Applications to engineering analysis of practical problems. Introduction to numerical analysis and visualization of flows.

Learning Resources, Technology & Texts

- **Main text:** F. A. Morrison, *An Introduction to Fluid Mechanics*, Cambridge Univ. Press, 2013. ISBN: 978-1-107-00353-8.
- **Additional readings:** None
- **Use of artificial intelligence (AI) or Large Language Models (LLM) in this course:** Generative AI platforms and LLMs are a technology that can be used to aid in writing and solving problems. The goals of this course are to increase your transport problem solving skills. Therefore, LLMs SHOULD NOT be used for any assignment unless otherwise noted. We will have at least one assignment where we explore the use of this tool together, but otherwise it should not be used as a short cut to complete assignments.
- **Brightspace learning management system:** Access the course via Purdue's Brightspace learning management system. Begin with the Start Here tab, which offers further insight to the course and how you can be successful in it. It is strongly suggested that you explore and become familiar not only with the site navigation, but also with the content and resources available for this course. See the Student Services widget on the campus homepage for resources such as Technology Help, Academic Help, Campus Resources, and Protect Purdue.

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Learning Outcomes

1. Explain the physical properties of a fluid and the consequences of such properties on fluid flow.
2. Apply momentum and/or force balances to the analysis of problems in hydrostatics.
3. Apply mass, momentum and energy balances for the solution of problems involving inviscid, steady flow.
4. Use appropriate macroscopic and microscopic mass, momentum, and energy balances to analyze steady laminar and turbulent flow of incompressible fluids in conduits and past submerged bodies.
5. Apply dimensional analysis and kinematic similarity for the interpretation of experimental fluid flow data and the design of fluid transfer equipment.
6. Utilize appropriate mass, momentum and energy balances for the design of fluid transfer equipment (pipelines, agitated vessels, pumps, etc.) and to analyze its operation.
7. Work professionally and ethically in teams to conduct fluid mechanics-based laboratory experiments.
8. Effectively report results in written form and practice safety as an integral part of laboratory work.

Program Outcomes for ABET

Graduates of the School of Chemical Engineering at Purdue University will (**bolded** items are addressed in this course):

1. **Apply principles of engineering, science, and mathematics to solve complex chemical engineering problems.**
2. **Apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.**
3. **Communicate effectively with a range of audiences.**
4. Recognize ethical and professional responsibilities in chemical engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.
5. **Function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.**
6. **Develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.**
7. **Acquire and apply new knowledge as needed, using appropriate learning strategies.**

Expectations

This is a 4 credit hour course, and it is expected that each student will spend a minimum of 12 hours each week outside of class working on homework, studying, and reading the course text (3 hours per credit). We encourage you to take studying seriously and establish good study habits (e.g., previewing the reading material before the lectures and practicing additional problems).

Instructors' Commitment

Your instructor will: (1) be courteous, punctual, well-organized, and prepared for lecture and other class activities; (2) answer questions clearly in class or arrange for detailed discussions out of class if in-class answers are not clear; (3) be available during office hours or notify you

beforehand if she is unable to keep to the original office hour schedule; and (4) grade uniformly and consistently to the posted guidelines.

Consulting with Faculty Members

We encourage you to discuss academic or personal questions during my office hours or via email. These discussions need not be limited to ChE 37700 content.

Assignments and Examinations

Problem sets

Homework assignments will be posted on Gradescope and will be due Wednesdays at 9pm. You will submit your homework assignment through Gradescope.

Assignments are intended to develop deeper understanding of concepts/techniques and their applicability to systems and processes of interest to the chemical engineer. Each problem is intended *to challenge you* to *extend and apply* concepts and techniques covered during lectures and recitations, *not* to simply practice "cookbook-type" solutions. Each assignment should be given serious attention. *Your diligence on the problem sets will pay dividends in helping develop understanding of complex concepts that you can demonstrate on exams* and ultimately take with you throughout your studies in chemical engineering.

- Homeworks will be assigned (in most cases) a week before the due date.
- Homeworks must be submitted *on the due date at the specified time*. Late assignments (without prior permission) *will not be accepted*, and will receive a zero.
- Your work should be prepared in a clear, organized, and well-documented fashion. It should be combined in a *single PDF* with *your name, problem set number, and due date* prominently displayed on the first page.
- I encourage you to work in groups, but be diligent about making sure that you are an active participant in the group and understand the concepts in each problem. In addition, so there are no questions about plagiarism and similar homework sets, *please list the members of the group you worked with for each assignment.*
- Homework assignments, solutions and grades will be posted on Brightspace and Gradescope.
- **The lowest grade from among the semester assignments will be dropped from the calculation of the course grade IF ALL assignments have been completed. For all assignments to be completed, you must have put in a full effort into the work you submit. *You will forfeit this benefit if you fail to satisfactorily complete and hand in one or more assignments*.**

Recitation participation

Each recitation period, 1-2 example problems will be covered by the teaching fellow. Students should work in small groups to solve these problems. The goal of recitation is to solve these problems within your group and, if necessary, with the assistance of the teaching fellow. The problems covered during these time periods will be similar to those used in homework assignments and exams. Attendance and professional behavior is expected at recitation.

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Exams

Announced exams will occur throughout the semester. The format of the exams will be closed book and closed notes. You will be allowed to make an equation sheet (a single sheet of paper) and an additional equation sheet will be provided to you with the exam. **Without prior approval from Professor Schultz, failure to complete any exam at the time that it is given will result in a zero for that exam.** No opportunities will be given to make up any previously unexcused exams.

Completed work will be submitted during the exam period on Gradescope AND the physical exam will be passed in at the end of the exam period. This ensures if a page is not scanned or not readable, we can find the physical exam and grade that page. Please be prepared for this and have a pdf converter on your phone, Gradescope will only accept pdfs. The time for submission will be IN ADDITION to the exam time, you will not have to use exam time to submit your work. This will differ for students with accommodations, so please contact Prof. Schultz about the different procedure.

Work on an exam will ONLY be accepted at the end of the exam period. Any work that has not been passed in by the end of the exam period WILL NOT be accepted under any circumstances.

Group Work and Study

By now, you have likely discovered your most effective learning style. It is important to recognize that there is great benefit that can be gained from personally working through the homework sets as well as discussing problem sets with your classmates. No matter which approach or combination of approaches you take, ***you*** *are responsible for working through and understanding each problem within a given problem set*. As such, *each student must hand in his or her* ***own work***. Verbatim copying of another classmate's homework set or previous homework solutions is strictly prohibited (see Plagiarism section below). If working in a group, please state the names of the people that you worked with at the top of the page. But as previously stated, *each individual is responsible for turning in their own work and not just a copy of another students work.*

Grading Scale

a. A range: 100 – 85% of the weighted points
b. B range: 84.9 – 75% of the weighted points
c. C range: 74.9 – 65% of the weighted points
d. D range: 64.9 – 55% of the weighted points
e. F: Less than 55% of the weighted points
f. If an exam or homework was too difficult (as judged by the faculty instructor) the grade may be scaled to a higher value. Grades will never be scaled downward. As a rule, scaling will not be applied. There is no preset distribution of final grades. The grading will reflect demonstrated student capability relative to an absolute performance standard that is expected of all Purdue chemical engineering students, rather than a scale that compares students to a mean performance metric on any evaluation vehicle. This means the entire class could receive any grade.

g. Lottery ticket A: ANY STUDENT WHO EARNS A GRADE OF 100% ON THE FINAL EXAM WILL AUTOMATICALLY RECEIVE A GRADE OF 'A' IN THE COURSE, INDEPENDENT OF EXAM AND LAB SCORES

Grade breakdown
Exam 1 and 2 : 20% each
Average of laboratory write – ups: 15%
Final exam (comprehensive): 30%
Homework: 13%
Participation: 2%
The participation grade will be determined from the level of participation of a student by the entire teaching team. Each student will start with 1 point and that can increase if there is an increased level of participation. This increased level can be consistent participation in class and/or recitation, coming to office hours, among other activities. This grade can also decrease if the student is unprofessional, disruptive or absent from most class activities.

Re-grading
Great effort will be made to carefully and fairly grade all homeworks, laboratory reports and exams. Should you feel that a mistake has been made on the grading of exams, please submit a *written request* that 1) specifies the problem requiring re-grading and 2) provides an explanation of the reasoning behind the request. This is only in the case of mistakes made during grading, comments on the grading rubrics and points assigned are not welcome. Re-grading **WILL NOT** be considered without a written request, which should include a scanned copy of the exam question and your answer. Unless otherwise noted, this must be submitted by the end of the day that the exams are returned. Please note, that the complete problem(s) will be re-graded, with no guarantee about the relative change in the grade.

Attendance Policy

This course follows the University Academic Regulations regarding class attendance, which state that students are expected to be present for every meeting of the classes in which they are enrolled. When conflicts or absences can be anticipated, such as for many University-sponsored activities and religious observations, you should inform the instructors of the situation as far in advance as possible. For unanticipated or emergency absences when advance notification to is not possible, contact the instructors as soon as possible by Purdue email or phone. For absences that do not fall under excused absence regulations (see below), this course follows the following procedures:

1. Do not come to class if you are feeling ill, but DO email the instructor at kmschultz@purdue.edu, with the subject line: ChE 377 absence. I do not need details about your symptoms. Just let me know you are feeling ill and cannot come to class. If it is an emergency, please follow the University regulations on medical care (see below).
2. Unless it falls under the University excused absence regulations (see below), any work due should be submitted on time via our course Brightspace or Gradescope.
3. If that day's class involves assessed work such as a test or presentation, you and the instructors will plan if and how you can make up the work, following the assignment guidelines. This plan

must be done before the next class period, so again, email the instructors immediately when you know that you will miss class.

4. The most important consideration in any absence is how it will affect your achievement of the assignment objectives and the course learning outcomes.

For cases that fall under **excused absence regulations**, you or your representative should contact or go to the Office of the Dean of Students (ODOS) website to complete appropriate forms for instructor notification. Under academic regulations, excused absences may be granted by ODOS for cases of grief/bereavement, military service, jury duty, parenting leave or certain types of medical care. The processes are detailed, so plan ahead.

Academic Integrity

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Academic integrity is one of the highest values that Purdue University holds. Individuals are encouraged to alert university officials to potential breaches of this value by either emailing integrity@purdue.edu or by calling 765-494-8778. While information may be submitted anonymously, the more information is submitted the greater the opportunity for the university to investigate the concern. More details are available on our course Brightspace under University Policies and Statements.

Academic dishonesty ***will not be tolerated*** in any form in this course. Specifically, Purdue prohibits "dishonesty in connection with any University activity. Cheating, plagiarism, or knowingly furnishing false information to the University are examples of dishonesty." [Section B-2-a, Code of Student Conduct] Furthermore, the University Senate has stipulated that "the commitment of acts of cheating, lying, and deceit in any of their diverse forms (such as the use of substitutes for taking examinations, the use of illegal cribs, plagiarism, and copying during examinations) is dishonest and must not be tolerated. Moreover, knowingly to aid and abet, directly or indirectly, other parties in committing dishonest acts is in itself dishonest." [University Senate Document 72-18, December 15, 1972] All incidents of academic dishonesty will be reported to the Dean of Students. **Such incidents include: (i) possessing or accessing, in hardcopy or electronic form, the solution manual to the course text, or to the exams (this includes consulting online sources or services which provide solutions that are obtained from the solutions manual to the course text); (ii) claiming credit for work that is not your own original work (for example, finding online solutions and copying and submitting them as your own work); and (iii) enabling other students to create work that is not their original work. The punishment for the first offense is a grade of zero for the entire work (exam or homework), and the punishment for a second offense is an F mark for the class.**

Use of copyrighted material

See the University Policies and Statements section of Brightspace for guidance on Use of Copyrighted Materials. Effective learning environments provide opportunities for students to reflect, explore new ideas, post opinions openly, and have the freedom to change those opinions over time. Students and instructors are the authors of the works they create in the learning environment. As authors, they own the copyright in their works subject only to the university's right to use those works for educational purposes Students may not copy,

reproduce, or post to any other outlet (e.g., YouTube, Facebook, or other open media sources or websites) any work in which they are not the sole or joint author or have not obtained the permission of the author(s).

Nondiscrimination Statement

Purdue University is committed to maintaining a community that recognizes and values the inherent worth and dignity of every person; fosters tolerance, sensitivity, understanding, and mutual respect among its members; and encourages each individual to strive to reach his or her potential. In pursuit of its goal of academic excellence, the University seeks to develop and nurture diversity. The University believes that diversity among its many members strengthens the institution, stimulates creativity, promotes the exchange of ideas, and enriches campus life. A hyperlink to Purdue's full Nondiscrimination Policy Statement is included in our course Brightspace under University Policies and Statements.

Accessibility

Purdue University strives to make learning experiences accessible to all participants. If you anticipate or experience physical or academic barriers based on disability, you are encouraged to contact the Disability Resource Center at: **drc@purdue.edu** or by phone: 765-494-1247, as soon as possible.

If the Disability Resource Center (DRC) has determined reasonable accommodations that you would like to utilize in my class, you must send me your Course Accommodation Letter. Instructions on sharing your Course Accommodation Letter can be found by visiting: **https://www.purdue.edu/drc/students/course-accommodation-letter.php** Additionally, you are strongly encouraged to contact me as soon as possible to discuss implementation of your accommodations.

Accommodations for Testing

Students with disabilities whose DRC Course Accommodation Letter (CAL) includes test accommodations must first release their CAL to me and then schedule to take their exams with Purdue Testing Services at https://www.purdue.edu/studentsuccess/testing-services/accommodated-testing/student.php. You must schedule at least four days (96 hours) before the exam date listed on the syllabus.

In the case of finals week, you must schedule by the Friday before quiet week. I will provide Purdue Testing Services with your exam and they will proctor it and provide the result to me for grading. Students must inform me immediately of cases when Purdue Testing Services is at capacity or otherwise unable to proctor the exam so that I can make other arrangements. Students who fail to follow this process and meet stated deadlines risk not being able to have their accommodations for that exam.

Mental Health/Wellness Statement

If you find yourself beginning to feel some stress, anxiety and/or feeling slightly overwhelmed, try Therapy Assistance Online (TAO), a web and app-based mental health resource available courtesy of Purdue Counseling and Psychological Services (CAPS)**.** TAO is available to all students at any time by creating an account on the TAO Connect website, or downloading the app from the App Store or Google Play. It offers free, confidential well-being resources through a self-guided program informed by psychotherapy research and strategies

that may aid in overcoming anxiety, depression and other concerns. It provides accessible and effective resources including short videos, brief exercises, and self-reflection tools.

If you need support and information about options and resources, please contact or see the Office of the Dean of Students. Call 765-494-1747. Hours of operation are M-F, 8 a.m.- 5 p.m.

If you find yourself struggling to find a healthy balance between academics, social life, stress, etc., sign up for free one-on-one virtual or in-person sessions in West Lafayette with a Purdue Wellness Coach at RecWell. Student coaches can help you navigate through barriers and challenges toward your goals throughout the semester. Sign up is free and can be done on BoilerConnect.

If you're struggling and need mental health services: Purdue University is committed to advancing the mental health and well-being of its students. If you or someone you know is feeling overwhelmed, depressed, and/or in need of mental health support, services are available. For help, such individuals should contact Counseling and Psychological Services (CAPS) at 765-494-6995 during and after hours, on weekends and holidays, or by going to the CAPS offices in West Lafayette.

Emergency Preparedness

In the event of a major campus emergency, course requirements, deadlines and grading percentages are subject to changes that may be necessitated by a revised semester calendar or other circumstances beyond the instructor's control. Relevant changes to this course will be posted onto the course website or can be obtained by contacting the instructors or TFs via email or phone. You are expected to read your @purdue.edu email on a frequent basis.

A link to Purdue's Information on Emergency Preparation and Planning is located on our Brightspace under "University Policies and Statements." This website covers topics such as Severe Weather Guidance, Emergency Plans, and a place to sign up for the Emergency Warning Notification System. I encourage you to download and review the *Emergency Preparedness for Classrooms document*.

The first day of class, we will review the **Emergency Preparedness plan for our specific classroom**, following Purdue's required Emergency Preparedness Briefing. Please make note of items like:

- The location to where we will proceed after evacuating the building if we hear a fire alarm.
- The location of our Shelter in Place in the event of a tornado warning.
- The location of our Shelter in Place in the event of an active threat such as a shooting.

Student Professionalism

The highest standards of professionalism and ethics are expected in CHE 37700. Each student is expected to come to class on time and not disrupt the class. Each student is also expected to behave in a professional manner and to follow Purdue's codes of student conduct, which includes the rights of students in violation of the code (http://www.purdue.edu/studentregulations/student_conduct/regulations.html). Each student is expected to exhibit consideration and respect towards the other students, the graders, the

teaching assistants (TAs), and the faculty members. Each student is expected to exhibit a positive attitude. Expectations for each student include (but are not limited to):

- Attending all class sessions.
- Coming to class and recitation on time and prepared by reading assigned material beforehand.
- Refraining from disrupting class (*e.g.*, turning off or silencing cell phones when they are not being used for class purposes and not carrying on a loud conversation during class).
- Maintaining the highest standards of academic honesty and integrity.
- Being an active contributor to team assignments.
- Being knowledgeable about the policies and information described in the syllabus.

Violent Behavior Policy

Purdue University is committed to providing a safe and secure campus environment for members of the University community. Purdue strives to create an educational environment for students and a work environment for employees that promote educational and career goals. Violent behavior impedes such goals. Therefore, violent behavior is prohibited in or on any University facility or while participating in any University activity.